

# Low-Light Image Enhancement with Cross-Dataset Generalization

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## Abstract

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## 1. Introduction

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## 2. Datasets and Splits

### 2.1. Datasets

The pipeline developed in this work is based on four datasets, each with a specific role in the experimental protocol: (i) LOL-v2 [4], (ii) LOL-v1 [3], (iii) MILLS [2], and (iv) ExDark [1]. All models are trained on a single source domain and are then evaluated on additional datasets without fine-tuning. This choice is intended to measure cross-dataset generalization, i.e., the ability of each model to enhance low-light images whose distribution differs from the training data.

The main training dataset is LOL-v2. This dataset contains both a real-captured subset, hereafter referred to as LOL-v2 Real, and a synthetic subset, referred to as LOL-v2 Synthetic. In the main protocol, only LOL-v2 Real is used, meaning that the low-light/normal-light pairs are acquired from real scenes rather than obtained through synthetic low-light degradation. LOL-v2 is a paired dataset: each low-light image is associated with a corresponding normally exposed reference image, which can be used as the target for supervised training and evaluation.

LOL-v1 is used only as a paired cross-dataset evaluation set. It represents a relatively close cross-dataset scenario, because it belongs to the same general benchmark family as LOL-v2.

MILLS is used as a second paired cross-dataset evaluation set. In particular, the low-resolution DSLR split, MILLS-DSLR, is adopted to keep the evaluation computationally manageable. Unlike LOL-v1, this dataset introduces a controlled illumination setting. Its input images are associated

with different illumination levels, and each input has a corresponding well-illuminated reference image. This makes it possible to evaluate the models not only with global results over the whole dataset, but also by studying how performance changes as the input illumination becomes more severe.

Finally, ExDark is used as an unpaired stress-test dataset. Here, unpaired means that the dataset provides low-light images without a corresponding normally exposed reference image. Therefore, ExDark is not used for reference-based comparison, but for no-reference evaluation and qualitative failure analysis on real-world dark scenes. Its category-based organization is preserved, meaning that images remain grouped according to the object category provided by the dataset, such as cars, people, animals, or indoor objects. This organization may help inspect whether some failure cases are related to the type of scene or object.

### 2.2. Dataset Licenses and Terms of Use

Since this project relies on publicly available datasets, their licenses and terms of use were reviewed to ensure that each dataset is used consistently with the conditions specified by its original authors.

For LOL-v1<sup>1</sup> and LOL-v2<sup>2</sup>, no explicit dataset license or terms of use were found in the official sources consulted and linked in this paper.

MILLS<sup>3</sup> is distributed through its Hugging Face dataset page, where the dataset metadata reports the license as MIT. This permissive license allows use, modification, and redistribution, provided that the copyright notice and permission notice are included in copies or substantial portions of the material.

ExDark<sup>4</sup> is subject to dataset-specific usage restrictions. Although the repository reports the project as open source under the BSD-3 license, it also states that commercial usage requires contacting the authors. More specifically, the dataset

<sup>1</sup>Official Source for LOL-v1

<sup>2</sup>Official Source for LOL-v2

<sup>3</sup>Official Source for MILLS

<sup>4</sup>Official Source for ExDark

Table 1. Dataset split sizes used in the pipeline. Counts refer to samples, i.e., low-light input images. For paired datasets, each input sample is associated with a corresponding target image.

| Dataset          | Train | Validation | Test |
|------------------|-------|------------|------|
| ExDark           | –     | –          | 7363 |
| LOL-v1           | –     | –          | 500  |
| LOL-v2 Real      | 620   | 69         | 100  |
| LOL-v2 Synthetic | 810   | 90         | 100  |
| MILLs            | –     | –          | 499  |

README indicates that the ExDark dataset may be used only for non-commercial research purposes.

All datasets are used exclusively within the scope of this non-commercial academic work. Any applicable license terms or usage restrictions are respected, and the corresponding dataset papers are cited in the bibliography.

To make these conditions explicit, a dedicated dataset notice is provided both in the project repository and in the accompanying release. The included datasets remain subject to the original authors’ licenses and terms of use, and are provided solely to support non-commercial academic reproducibility.

### 2.3. Split Construction

Split construction follows the role assigned to each dataset, as discussed in Section 2.1. For the main source domain, LOL-v2 Real, the official training split is used to derive the training and validation partitions. In particular, a fixed fraction of the official training samples is deterministically held out as validation data using a fixed random seed. The official test split is kept untouched for the final in-domain evaluation. In this context, “in-domain” means that training, validation, and testing images come from the same dataset and acquisition setting.

The synthetic subset of LOL-v2 is kept separate from the main protocol and is used only for an additional ablation study, aimed at evaluating whether synthetically degraded low-light examples improve or degrade generalization. When this subset is used, a validation partition is deterministically derived from the official training split using the same procedure adopted for LOL-v2 Real, ensuring consistency in the split construction process.

For the remaining datasets, no additional train or validation partitions are created. Instead, all available samples are assigned to a single evaluation partition. For datasets that already provide predefined splits, such as LOL-v1 and MILLs, the original partitions are merged into a unified test set. ExDark is likewise converted into a single test split, while preserving the original category labels as metadata.

Following the split construction procedure, the resulting split sizes are summarized in Table 1.

## 3. Experimental Protocol

### 3.1. General Strategy

The experimental protocol is designed to prevent test leakage and to ensure that the final evaluation of the studied models reflects genuine generalization performance. The main goal of the project is not only to obtain good in-domain enhancement performance on the main dataset, but also to understand whether a model trained on this source domain can generalize to different low-light datasets without fine-tuning.

To address this goal, the project is organized around model families. For each model family, the experimental protocol is kept fixed and a dedicated grid search is defined to explore meaningful architectural and training choices. Whenever possible, all the configurations share the same experimental conditions, including the early stopping rule and the data augmentation policy. The remaining hyperparameters are explored through a grid whose size is kept comparable across model families, so that different families are compared under a similar experimental budget.

All models are trained on a single source domain, LOL-v2 Real. The training split is used to optimize the model parameters, while the validation split is used for early stopping and model selection. The test sets of all datasets are never used during these steps. They are evaluated only after the validation-based selection procedure has been completed, in order to provide the final evidence of model performance and generalization.

For each model family, the experimental protocol is organized into four main stages: (i) a single-seed screening stage, in which all configurations of the grid search are trained with a single seed and ranked using validation performance; (ii) a multi-seed re-training stage, in which the three most promising configurations are trained with multiple random seeds to estimate both performance and stability; (iii) a validation-based model selection stage, in which the final configuration is chosen from the aggregated multi-seed validation results; and (iv) a final evaluation stage, in which the selected configuration is tested on the in-domain test set and on all cross-dataset test sets. Figure 1 provides a visual overview of this workflow. The following subsections describe each stage in detail.

### 3.2. Single-Seed Screening

The first experimental stage is a single-seed grid search, in which every configuration is trained using a single fixed seed. The purpose of this stage is not to produce final evidence, because a single seed can be affected by random initialization and data-loader randomness. Instead, the purpose is to reduce the search space and identify a small number of promising configurations.

Each configuration is trained on the LOL-v2 Real training

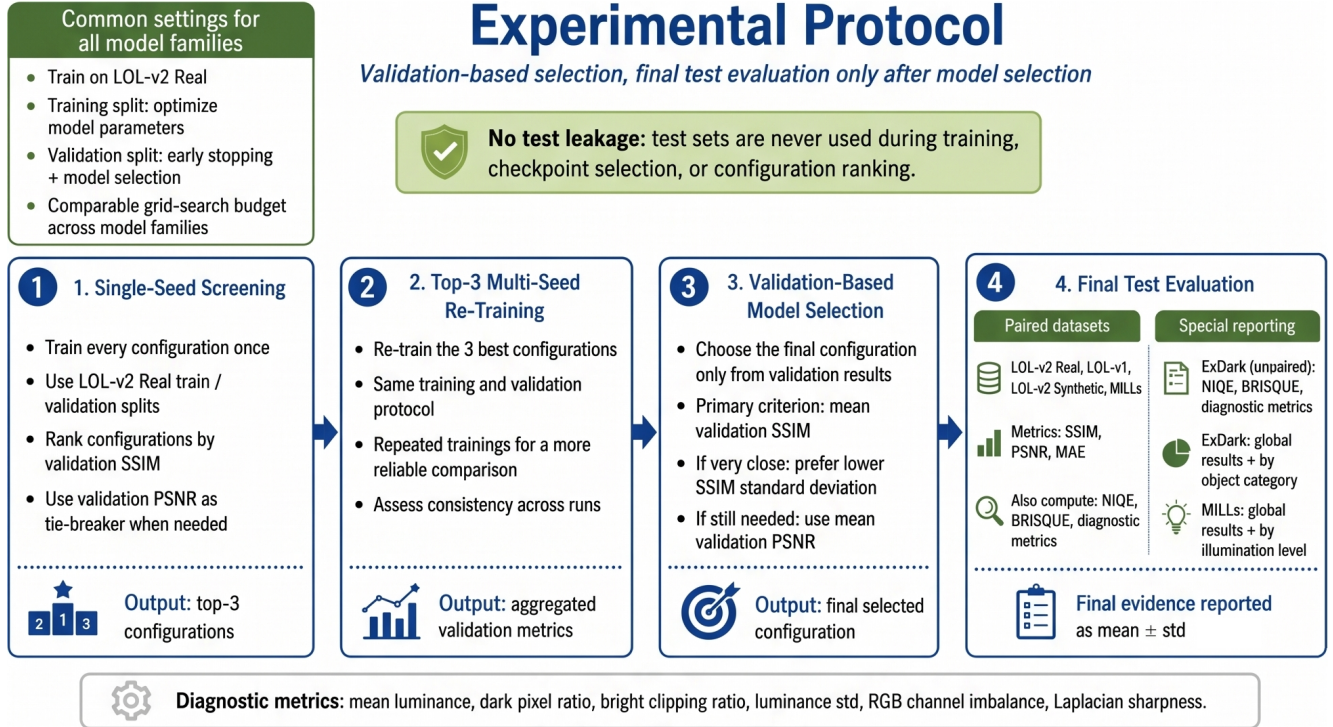


Figure 1. Overview of the experimental protocol.

split and monitored on the LOL-v2 Real validation split. The test sets are not evaluated during screening. For each run, validation structural similarity index measure (SSIM) is used as the primary criterion for checkpoint selection. This choice reflects the goal of low-light enhancement: the output should remain structurally similar to the reference image, preserving luminance, contrast, and local image structure, rather than only reduce the average pixel-wise error. Validation peak signal-to-noise ratio (PSNR) is also computed and used as a tie-breaker because it measures pixel-wise reconstruction quality in logarithmic form. Therefore, when two checkpoints achieve validation SSIM values within the configured tolerance, PSNR is used to select the best checkpoint for that run.

The output of this stage is a validation-based ranking of the explored configurations. This ranking is later parsed from the run logs and used to select the top-3 configurations for the multi-seed stage.

### 3.3. Top-3 Multi-Seed Re-Training

After the screening stage, the three most promising configurations are selected from the validation-based ranking and re-trained under the same training and validation protocol. This stage is necessary because the screening phase provides only a preliminary estimate of performance, obtained from a single training run for each configuration.

The goal of the multi-seed stage is to obtain a more re-

liable comparison among the selected configurations. By repeating the training procedure several times for each of them, the protocol can evaluate whether its validation performance is consistent or mainly due to a favorable run.

The outputs of this stage are aggregated validation metrics for each selected configuration. These aggregated results are then used for the final validation-based selection, while the test sets remain untouched.

### 3.4. Validation-Based Model Selection

The final configuration is selected only from the aggregated validation results of the multi-seed stage.

The primary selection criterion is the mean validation SSIM across runs. In this case, where configurations obtain very close mean SSIM values, the selection favors the more stable configuration, measured by the standard deviation of validation SSIM. If the comparison is still ambiguous, mean validation PSNR is used as a final tie-breaker.

This rule keeps the whole selection process validation-based, while also accounting for statistical robustness across repeated trainings.

### 3.5. Final Test Evaluation

The final test evaluation is performed only after the final configuration has been selected. The selected configuration is evaluated on the test set of all the datasets used in this project, and the resulting metrics are aggregated as mean and



standard deviation. This ensures that the final evidence is not based on a single run, but reflects both average performance and statistical robustness.

The evaluation protocol depends on whether a dataset is paired or unpaired. In paired datasets, “full-reference” metrics can be computed by comparing the enhanced output of the model with the corresponding target image of the dataset. In this work, paired datasets are evaluated using SSIM, PSNR, and mean absolute error (MAE). This last metric measures the average absolute pixel difference between the enhanced image and the reference image.

“No-reference” metrics do not require target images and are computed for both paired and unpaired datasets. In this work, the no-reference metrics are naturalness image quality evaluator (NIQE) and blind/referenceless image spatial quality evaluator (BRISQUE). More specifically, NIQE measures the deviation of an image from statistical properties commonly observed in natural images, while BRISQUE estimates perceptual quality from spatial scene statistics.

Additional diagnostic metrics are computed on the model output for both paired and unpaired datasets. They are used to support the interpretation of failure modes, meaning recurring types of enhancement errors. These diagnostics include: (i) mean luminance, which measures the average brightness of the enhanced image; (ii) dark pixel ratio, which measures the fraction of pixels that remain very dark; (iii) bright clipping ratio, which measures the fraction of pixels close to saturation; (iv) luminance standard deviation, which provides a compact indication of contrast variation; (v) RGB channel imbalance, which helps detect color cast; and (vi) Laplacian sharpness, which provides a simple estimate of high-frequency content related to blur, noise, or halo artifacts.

LOL-v1 and LOL-v2 are paired datasets and are therefore evaluated by aggregating full-reference metrics, no-reference metrics, and diagnostic output metrics over their respective test sets. ExDark and MILs require a more specialized reporting protocol because of their particular dataset structure and evaluation objectives.

For ExDark, which is unpaired, no normally exposed target image is available. Therefore, it is evaluated using only no-reference metrics and diagnostic output metrics. In addition to global results over the whole dataset, ExDark is also analyzed by object category, since its images are organized into semantic groups.

MILs is a paired dataset, so it is evaluated like LOL-v1 and LOL-v2, but the results are reported at two levels: globally over all samples and grouped by illumination level. This grouped reporting is useful because MILs contains inputs acquired at different light intensities, making it possible to analyze how model performance changes across illumination levels.

## 4. End-to-End Pipeline

### 4.1. Overview of the Implemented Workflow

The pipeline connects the datasets and splits discussed in Section 2 with the experimental protocol defined in Section 3. Its role is to transform raw datasets into a common representation and then use that representation to train, select, evaluate, and analyze the models in a reproducible way.

The implemented workflow is organized into five main stages: (i) manifest generation, where raw dataset folders are converted into CSV manifests; (ii) offline preprocessing, where images are deterministically processed once, updated manifests are written, and dataset-level statistics are computed; (iii) experiment execution, where configuration files are expanded into concrete runs and the framework performs online data augmentation, training, validation-based model selection, checkpointing, logging, and optional test evaluation; (iv) log parsing and aggregation, where the resulting JSONL logs are used to build rankings and aggregate validation or test metrics; and (v) qualitative example export, where selected checkpoints are restored to produce visual examples for inspection.

Figure 2 summarizes the implemented workflow and shows how the dataset preparation stages are connected to experiment execution, log aggregation, and qualitative analysis.

“Offline” preprocessing and “online” augmentation serve different purposes in the pipeline. Offline preprocessing consists of deterministic operations that are performed once before training, saved on disk, and kept fixed across experiments. Online augmentation is instead applied at training time, when samples are loaded from the training split, and has a stochastic nature because the transformations are randomly sampled during training. In the current configuration, geometric augmentations are applied consistently to both the low-light input and the paired target, while photometric perturbations are applied only to the input image. This preserves the alignment of paired samples while exposing the model to additional low-light variations.

The pipeline follows a central design principle of the project: only the manifest generation stage is intentionally dataset-specific. This stage absorbs the differences between the original dataset structures, such as pairing rules, filename conventions, category folders, and illumination-level metadata. Once each dataset has been converted into the CSV manifest format, the following stages of the pipeline operate on the same representation and do not need to know how each dataset was originally organized on disk.

The following subsections describe these stages in more detail, showing how each part of the pipeline is implemented in the codebase.

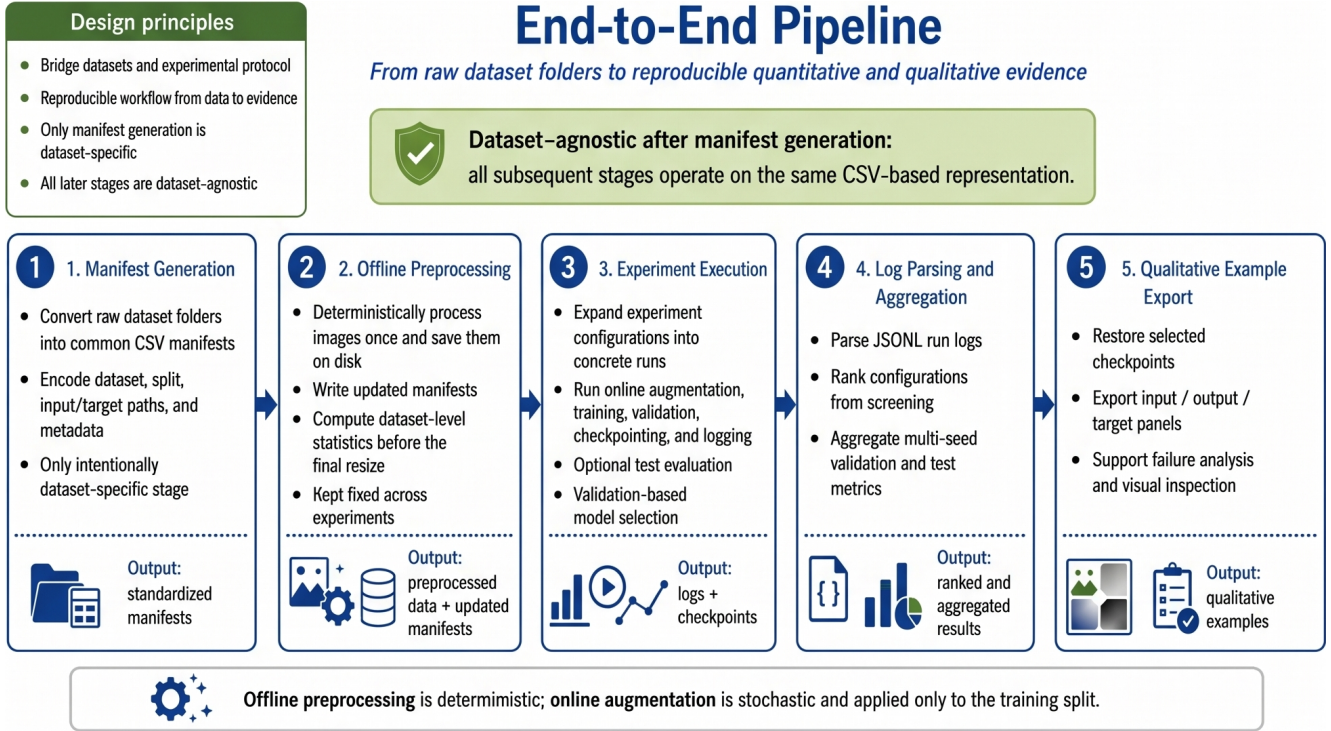


Figure 2. Overview of the pipeline.

## 4.2. Manifest Generation

The purpose of the first stage of the pipeline is to scan the original dataset folders and build CSV manifest files from their structure. In each manifest, every row represents one image sample and stores the dataset name, the split to which the sample belongs, the path of the low-light input image, the path of the corresponding target image when available, and optional metadata such as the image category or the illumination level.

This step is necessary because the datasets used in this project are not organized in the same way. Some datasets provide paired low-light and normal-light images with the same filename, while others use different filename conventions. ExDark is instead unpaired and organized by object category. MILs also includes an illumination level in the input filename, which is useful for analyzing how performance changes under different lighting conditions. Without this standardization step, the subsequent scripts in the pipeline would need to contain several dataset-specific rules.

For this reason, manifest generation is the only intentionally dataset-specific part of the pipeline. The script reads a YAML configuration file that describes where each dataset is located, which output file should be produced, and how the corresponding manifest should be created. In LOL-v1 and LOL-v2 Synthetic, low-light images and their corresponding reference images share the same filename, making the asso-

ciation straightforward. In LOL-v2 Real, the filenames differ only because low-light and reference images use different prefixes. ExDark follows a completely different structure: images are grouped by category and no reference image is provided, so only the image path and category information are recorded. Finally, in MILs, the filename also contains information about the illumination level, which is extracted and stored together with the path of the corresponding reference image.

The output of this step is a CSV manifest file for each dataset, with a fixed column structure. This common format is what makes the later stages dataset-agnostic: once the manifests have been created, preprocessing, training, and evaluation only need to read the paths and metadata stored in the CSV files, without knowing how the original dataset was arranged on disk.

The manifest generation script also supports reproducibility-oriented choices. When required by the protocol and when a training split is available, a validation split can be sampled from the training split using a fixed random seed. Moreover, the generated rows are sorted according to the split and image path, making the row order deterministic across runs.

The same manifest information is also useful during evaluation. For ExDark, the object category associated with each image makes it possible to report results not only globally, but also separately for each category. For MILs, the illu-

mination level associated with each image makes it possible to report results both over the whole dataset and separately for each illumination level. In this way, the evaluation can remain general while still preserving the specific information needed for each dataset.

### 4.3. Offline Preprocessing

After manifest generation, all datasets are processed by an offline preprocessing script. This stage is called “offline” because it contains deterministic preliminary operations that are performed once for all the experiments of this project rather than repeatedly during training or evaluation. The processed images are stored on disk and are then used by all later stages of the pipeline.

The script reads the CSV manifests produced in the previous stage and creates a preprocessed version of each dataset on disk. For each dataset, it also writes a new CSV manifest with the same columns as the original one, but with paths updated to the preprocessed images. The output paths preserve the dataset-relative folder structure of the original files, while the image extension is replaced according to the selected output format.

Each image undergoes the following deterministic operations: (i) loading with EXIF orientation applied when available, (ii) conversion to RGB, (iii) resizing to  $256 \times 256$  using Lanczos interpolation, and (iv) saving in PNG format. Applying the EXIF orientation avoids processing photographs with an incorrect rotation. PNG is used to avoid introducing additional compression artifacts during preprocessing. If the same source image is referenced more than once in a manifest, it is processed only once and then reused, avoiding possible duplicated work.

A fixed resizing strategy is used instead of cropping. In this project, preserving the whole scene during preprocessing was considered preferable to selecting a local region, because a crop would permanently discard part of the image and could remove relevant low-light regions, light sources, or objects needed for cross-dataset analysis. Although resizing may slightly distort the aspect ratio of some images, it keeps the full scene and provides a fixed deterministic input size for all datasets. This choice does not exclude the use of cropping as online data augmentation during training: unlike offline cropping, random resized crops are stochastic, do not permanently modify the preprocessed dataset, and are applied only to the training split, while validation and test images remain deterministic.

Before processing a manifest, the script performs a lightweight consistency check. It verifies that no input path is duplicated within the same split and that the same input path does not appear in different splits. These checks are intentionally limited, since the dataset-specific pairing rules have already been handled during manifest generation.

In addition to producing the preprocessed images and

updated manifests, the script computes simple dataset-level statistics. These statistics are computed before applying the final resize, after loading the image with the correct orientation and converting it to RGB. Therefore, they summarize the illumination properties of the original images rather than only the  $256 \times 256$  images later used as model inputs.

For each image of size  $H \times W$ , a luminance map  $Y \in [0, 1]^{H \times W}$  is computed from the RGB channels using the ITU-R BT.709 luma coefficients:

$$Y_{i,j} = 0.2126R_{i,j} + 0.7152G_{i,j} + 0.0722B_{i,j}, \quad (1)$$

where  $R_{i,j}$ ,  $G_{i,j}$ , and  $B_{i,j}$  are the RGB values of pixel  $(i, j)$ , scaled to  $[0, 1]$ . Based on this luminance map, the dark pixel ratio is defined as

$$\text{DPR} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W 1[Y_{i,j} < \tau_d], \quad \tau_d = 0.10, \quad (2)$$

where  $H$  and  $W$  are the image height and width respectively,  $\tau_d$  is the dark-pixel threshold, and  $1[\cdot]$  is the indicator function, equal to one when the condition is true and zero otherwise.

For each dataset, the script records the number of samples, the median original input size, the mean and standard deviation of input luminance, and the average fraction of dark pixels. For paired datasets, the same luminance summary is also computed on the target images. These statistics are written to a dataset summary file and are reported in Table 2.

The statistics in Table 2 provide a compact description of the illumination distribution encountered by the preprocessing pipeline. LOL-v1 and LOL-v2 Real have very similar input luminance distributions, supporting the use of LOL-v1 as a close cross-dataset benchmark for the main training domain. LOL-v2 Synthetic and MILs are instead brighter on average, indicating a larger distribution shift in terms of illumination. ExDark lies between these cases in terms of input luminance and remains useful as an unpaired real-world stress test.

## 5. Conclusion

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## References

- [1] Yuen Peng Loh and Chee Seng Chan. Getting to know low-light images with the exclusively dark dataset. *Comput. Vis. Image Underst.*, 178:30–42, 2019. 1
- [2] Maria Pilligua, David Serrano-Lozano, Pai Peng, Ramon Baldrich, Michael S. Brown, and Javier Vazquez-Corral. Evaluating low-light image enhancement across multiple intensity levels. *CoRR*, abs/2511.15496, 2025. 1
- [3] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Deep retinex decomposition for low-light enhancement. In *BMVC*, page 155. BMVA Press, 2018. 1

Table 2. Dataset statistics computed from the preprocessed datasets before the final resize. Luminance values are reported as mean  $\pm$  standard deviation across samples. Dark pixels denote the average fraction of input pixels with luminance below 0.10.

| Dataset          | Samples | Median original size | Input luminance   | Dark pixels | Target luminance  |
|------------------|---------|----------------------|-------------------|-------------|-------------------|
| ExDark           | 7363    | $640 \times 500$     | $0.143 \pm 0.086$ | 59.6%       | –                 |
| LOL-v1           | 500     | $600 \times 400$     | $0.061 \pm 0.041$ | 82.7%       | $0.458 \pm 0.102$ |
| LOL-v2 Real      | 789     | $600 \times 400$     | $0.062 \pm 0.042$ | 80.9%       | $0.436 \pm 0.105$ |
| LOL-v2 Synthetic | 1000    | $384 \times 384$     | $0.240 \pm 0.138$ | 46.7%       | $0.513 \pm 0.141$ |
| MILLs            | 499     | $600 \times 400$     | $0.244 \pm 0.110$ | 16.4%       | $0.369 \pm 0.095$ |

- [4] Wenhan Yang, Wenjing Wang, Haofeng Huang, Shiqi Wang, and Jiaying Liu. Sparse gradient regularized deep retinex network for robust low-light image enhancement. *IEEE Trans. Image Process.*, 30:2072–2086, 2021. [1](#)